

TRANSPORT AND TILING BOUNDS FOR THE AFFINE PLANK PROBLEM ON THE TRIANGLE

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ABSTRACT. Bang’s affine (relative-width) plank conjecture asserts that if a convex body K is covered by planks $P_1, \dots, P_n$ then $\sum_i \text{rw}_K(P_i) \geq 1$, where $\text{rw}_K(P) = w(P)/w_K(u_P)$ is the width of P relative to the width of K in P ’s direction. It is open for non-symmetric bodies, and open already for three planks in the plane. We study the triangle as a concrete test body. We give (i) a short single-measure proof of $\sum \text{rw} \geq 1/d$ for $d \geq 2$, whose per-plank estimate is sharp at the simplex, so that no single-measure argument can improve on $1/d$; (ii) sharp $\sum \text{rw} \geq 1$ for two directions and for facet-parallel families (all dimensions); (iii) a Hunter-free proof of the “three facets + one arbitrary plank” case; and, as our main results, (iv) what is, to our knowledge, the first *tight non-facet* case with full rigidity: three planks parallel to the medians of a triangle satisfy $\sum \text{rw} \geq 1$ with equality if and only if each is the central third of its median; (v) a *weighted-perimeter theorem* extending the bound to a two-parameter family of generic concurrent direction triples, via an explicit witness measure with edge masses $1 - 2p_j$; and (vi) tightness and rigidity across that entire family: the bound is attained at every member by an explicit covering, unique for its directions — a two-parameter family of sharp configurations. We isolate the transport method behind (i)–(vi) as a *uniform-marginal witness measure* and give a dimension-agnostic necessary condition for its existence (concurrence of the mid-level lines); on the triangle the condition is also sufficient, giving a complete characterization for three directions: a witness measure exists if and only if the mid-level lines concur. Finally we record a structural obstruction: the chord-lemma step in the sign argument of Bang, which yields the chord-normalized bound $\sum w(P_i)/\ell_K(u_i) \geq 1$, is sharp, so that argument cannot be pushed past the longest chord: no normalizer strictly between chord and width is reachable by it. We claim no proof of the conjecture.

1. INTRODUCTION

A *plank* of width w is the region between two parallel hyperplanes at distance w . For a convex body $K \subset \mathbb{R}^d$ and a plank P with unit normal u , the *relative width* of P is $\text{rw}_K(P) = w(P)/w_K(u)$, where $w_K(u) = \max_{x \in K} \langle u, x \rangle - \min_{x \in K} \langle u, x \rangle$. Bang’s affine plank conjecture [3, 1] states:

Conjecture 1.1 (Affine plank problem; Bang 1951). If K is covered by planks $P_1, \dots, P_n$, then $\sum_{i=1}^n \text{rw}_K(P_i) \geq 1$.

Ball [2] proved Conjecture 1.1 for centrally symmetric K . For general bodies it is open, and, as emphasised by Bakaev–Yehudayoff [4], it is open *already in the plane* — only the two-plank case is settled there. The best general lower bound is $\sum \text{rw} \geq 2/(1 + \sqrt{d})$ [4]; on the equilateral triangle their method gives $\sum \text{rw} \geq 4\sqrt{3} - 6 \approx 0.928$.

Scope. This paper studies the triangle as a concrete body and develops the associated *transport* (witness-measure) and *tiling* methods. It is *not* an attack on Conjecture 1.1 through the Ambrus reduction: that reduction sends a covering by N planks to a simplex of dimension $2N - 1$, so the triangle (the 2-simplex) is never a target (Remark 1.2). Our results are new sharp cases

for a fixed body, a rigidity theorem, a dimension-agnostic structural characterization, and an obstruction; we prove nothing about the general conjecture.

Remark 1.2. Ambrus’s reduction [1] sends a covering by N planks to a covering of a simplex of dimension $2N - 1$ (two points per plank). The target simplices thus have odd dimension growing with N ; the 2-simplex (triangle) is never among them. So proving the conjecture for the triangle does *not* imply the planar case, and our results are about the triangle as a concrete body, not a reduction of Conjecture 1.1.

The triangle model. By affine invariance take $\Delta = \{x \in \mathbb{R}^3 : x_i \geq 0, x_1 + x_2 + x_3 = 1\}$. A normalized plank is $P = \{x \in \Delta : f(x) \in I\}$ with $f : \Delta \rightarrow [0, 1]$ affine onto $[0, 1]$ and $I \subset [0, 1]$ an interval; then $\text{rw}_\Delta(P) = |I|$. Covering means $\Delta \subset \bigcup_i P_i$. The restriction $I \subset [0, 1]$ loses no generality: replacing I by $I \cap [0, 1]$ leaves $P \cap \Delta$ unchanged and does not increase $|I|$.

Results. Theorem 2.1 (§2), Theorems 3.1 and 3.2 (§3), Theorem 4.1 (§4), and the centerpieces: Theorem 5.1 with rigidity (§5); the weighted-perimeter Theorem 6.3 (§6), which extends the sharp bound from the medians to a two-parameter family of generic concurrent triples; and Theorem 6.6, which shows the bound is attained at every member of that family by an explicit unique tight covering. Proposition 6.1 (§6) is the dimension-agnostic necessary condition, Corollary 6.4 the resulting characterization for cyclic triples, and Theorem 6.8 the full three-direction characterization on the triangle; Proposition 7.1 (§7) is the normalizer obstruction. Appendix A classifies the boundary tilings of cyclic triples (Proposition A.1), the combinatorial engine behind both rigidity results.

2. THE SINGLE-MEASURE BOUND $\sum \text{rw} \geq 1/d$

Theorem 2.1. *Let $d \geq 2$. If $K \subset \mathbb{R}^d$ is covered by planks $P_1, \dots, P_n$, then $\sum_i \text{rw}_K(P_i) \geq 1/d$. The per-plank estimate $\mu(P) \leq d \text{rw}_K(P)$ used in the proof is sharp at the simplex; hence no single-measure argument can improve the constant $1/d$. (For $d = 1$ the conjecture itself is trivial: intervals covering an interval satisfy $\sum \text{rw} \geq 1$.)*

Proof. Let μ be the uniform probability measure on K . Fix u , normalize $\langle u, \cdot \rangle \in [0, w]$, $w = w_K(u)$. The marginal density is $f(t) = V(t)/\text{vol}(K)$ with $V(t) = \text{vol}_{d-1}(K \cap \{\langle u, \cdot \rangle = t\})$; by Brunn–Minkowski $g := V^{1/(d-1)}$ is concave on $[0, w]$. If g peaks at t^* with value g^* , then $g \geq g^*h$ for the tent h ($h(0) = h(w) = 0$, $h(t^*) = 1$), so $f = g^{d-1} \geq (\max f) h^{d-1}$ and $1 = \int f \geq (\max f) \int_0^w h^{d-1} = (\max f) w/d$, giving $\max f \leq d/w$. Hence $\mu(P) = \int_I f \leq d \text{rw}_K(P)$ for each plank, and $1 = \mu(K) \leq \sum_i \mu(P_i) \leq d \sum_i \text{rw}_K(P_i)$. For sharpness of the per-plank estimate: equality in $\max f \leq d/w$ forces $V \propto h^{d-1}$ (conical sections), realized by the simplex $K = \Delta^d$ with u a vertex–facet direction, where a thin plank at the facet has $\mu(P) = (d - o(1)) \text{rw}_K(P)$. Thus for the simplex no single probability measure can certify a constant larger than $1/d$ by the union bound; this does *not* assert any covering of the simplex with $\sum \text{rw}$ near $1/d$ (indeed Theorem 3.2 gives $\sum \text{rw} \geq 1$ for facet-parallel coverings). $\square$

Remark 2.2. The bound cannot exceed $1/d$ by a single measure: uniform marginals in all directions would force $\mathbb{E}[x_k] = 1/2$ against $\sum x_k = 1$. This is Gardner’s obstruction [5]: for the triangle there is no relative-width measure for the three side-directions.

3. SHARP CASES: TWO DIRECTIONS AND FACET-PARALLEL FAMILIES

Theorem 3.1. *If every plank covering the triangle is parallel to one of two fixed non-parallel directions, then $\sum \text{rw} \geq 1$, and this is sharp. (If the two directions are parallel the statement is trivial: all planks share one normalized coordinate f , the intervals $f(P_i)$ cover $[0, 1]$, and $\sum \text{rw} = \sum |I_i| \geq 1$.)*

Proof. Let $f_u, f_v : \Delta \rightarrow [0, 1]$ be the two affine coordinates; since the directions are non-parallel, $\Phi = (f_u, f_v)$ is an affine bijection onto a triangle $T \subset [0, 1]^2$ with full projections. By [5, Theorem 1], which constructs a relative-width measure for any *two* directions on any planar convex body, there is a probability measure ν on Δ with $f_{u\#}\nu = f_{v\#}\nu = \text{Leb}$. Then $\nu(P) = |I| = \text{rw}(P)$ for each plank, and $1 = \nu(\Delta) \leq \sum \nu(P_i) = \sum \text{rw}(P_i)$. $\square$

Theorem 3.2. *Planks parallel to the facets of a simplex Δ^d , in any number, that cover Δ^d satisfy $\sum \text{rw} \geq 1$, and this is sharp.*

Proof. Write $\Delta^d = \{\lambda \in [0, 1]^{d+1} : \sum_k \lambda_k = 1\}$ in barycentric coordinates; a plank parallel to facet k is a coordinate slab $\{\lambda_k \in I\}$. Grouping the planks by facet, let $U_k = \bigcup_{C(i)=k} I_i \subset [0, 1]$ be the forbidden set on facet k , so $\sum_k |U_k| \leq \sum_i |I_i| =: R$, and set $C_k = [0, 1] \setminus U_k$. A point $\lambda \in \Delta^d$ is uncovered iff $\lambda_k \in C_k$ for every k ; since $C_k \subset [0, 1]$, an uncovered point exists iff $1 \in C_1 + \dots + C_{d+1}$ (Minkowski sum).

Assume $R < 1$; we produce an uncovered point. Each C_k is nonempty compact ($|C_k| = 1 - |U_k| \geq 1 - R > 0$). Put $D = C_1 + \dots + C_d$. By the one-dimensional Brunn–Minkowski inequality $|A+B| \geq |A|+|B|$ (for nonempty compact $A, B \subset \mathbb{R}$, iterated), $|D| \geq \sum_{k=1}^d |C_k| = d - \sum_{k=1}^d |U_k|$; as $D \subseteq [0, d]$, $|[0, 1] \setminus D| \leq |[0, d] \setminus D| \leq \sum_{k=1}^d |U_k|$, whence $|D \cap [0, 1]| \geq 1 - \sum_{k=1}^d |U_k|$. Also $(1 - C_{d+1}) \cap [0, 1]$ has measure $|C_{d+1}| = 1 - |U_{d+1}|$. Both sets lie in $[0, 1]$ and their measures sum to $2 - \sum_k |U_k| \geq 2 - R > 1$, so they intersect: there is $t \in D$ with $1 - t \in C_{d+1}$. Then $t + (1 - t) = 1 \in C_1 + \dots + C_{d+1}$, i.e. an uncovered λ exists. Contrapositively, covering forces $R \geq 1$. Sharpness: the facet tiling $I_k = [0, \frac{1}{d+1}]$ gives $\sum \text{rw} = 1$. $\square$

Remark 3.3. The edge-by-edge argument does *not* prove this: on an edge only two barycentric coordinates vary, so no single edge can force $\sum_k |U_k| \geq 1$ across all $d+1$ facets. The Minkowski-sum reduction above is the correct route; note that no witness *measure* can prove it either, since facet directions violate the concurrence condition of Proposition 6.1.

4. THREE FACETS PLUS ONE ARBITRARY PLANK

Theorem 4.1. *Let $P_i = \{\lambda_i \in I_i\}$ ($i = 1, 2, 3$) be facet-parallel with $a_i = |I_i|$, and let $Q = \{f \in J\}$, $t = |J|$, with $f : \Delta \rightarrow [0, 1]$ affine normalized. If $\Delta \subset P_1 \cup P_2 \cup P_3 \cup Q$ then $a_1 + a_2 + a_3 + t \geq 1$.*

Proof. Choose an active edge of f ; relabelling, $f(v_j) = 0$, $f(v_k) = 1$, and let v_i be the opposite vertex, $\theta = f(v_i)$. For $s \in [0, 1]$ consider the fibre $F_s = \{\lambda_i = s\}$, parametrized by $u = \lambda_k \in [0, 1 - s]$, so $\lambda_j = 1 - s - u$ and $f|_{F_s}(u) = \theta s + u$. On F_s : $P_i \cap F_s = F_s$ if $s \in I_i$ (else $\emptyset$); $P_j \cap F_s$, $P_k \cap F_s$, $Q \cap F_s$ have lengths $\leq a_j, a_k, t$. Put $B = a_j + a_k + t$. If $s \notin I_i$ and F_s is covered, its length $1 - s \leq B$. If $B \geq 1$ we are done; else $[0, 1 - B] \subset I_i$, so $a_i \geq 1 - B$. In both cases $a_1 + a_2 + a_3 + t = a_i + B \geq 1$. $\square$

5. THE MEDIAN THEOREM AND ITS RIGIDITY

Let m_1, m_2, m_3 be the medians of Δ as normalized affine forms: with A, B, C the vertices,

$$m_1 = (\tfrac{1}{2}, 0, 1), \quad m_2 = (0, 1, \tfrac{1}{2}), \quad m_3 = (1, \tfrac{1}{2}, 0),$$

(vertex values). One has $m_1 + m_2 + m_3 \equiv \frac{3}{2}$. A *median plank* is $P_i = \{m_i \in I_i\}$, $I_i = [l_i, h_i]$, $r_i = |I_i| > 0$.

Theorem 5.1. *If three median planks cover Δ , then $r_1 + r_2 + r_3 \geq 1$, with equality if and only if $I_1 = I_2 = I_3 = [\frac{1}{3}, \frac{2}{3}]$.*

To our knowledge this is the first tight *non-facet* case of the three-plank problem with full rigidity: the sharp cases recorded in the survey [6] (and in [5, 4]) concern facet/coordinate

directions, two directions, or symmetric bodies, and we are not aware of a comparable tilted tight instance in the literature.

Proof of the bound. Let μ be the perimeter measure: uniform in the parameter on each of the three edges, mass $\frac{1}{3}$ each. A direct computation shows each pushforward $m_{i\#}\mu = \text{Leb}[0, 1]$. Indeed for m_1 : on AB , m_1 ranges over $[0, \frac{1}{2}]$ with Jacobian 2 (density $\frac{1}{3} \cdot 2$); on CA , over $[\frac{1}{2}, 1]$ likewise; on BC , over $[0, 1]$ with Jacobian 1 (density $\frac{1}{3}$); the total density is 1 on $[0, 1]$. Hence $\mu(P_i) = \text{Leb}(I_i) = r_i$, and $1 = \mu(\Delta) \leq \sum \mu(P_i) \leq \sum r_i$. $\square$

Proof of rigidity. Equality in $1 \leq \sum \mu(P_i)$ forces the P_i to partition μ a.e.; since μ lives on the edges, on each edge the three traces tile $[0, 1]$. We claim there are exactly three such edge-tilings, and only one covers the interior.

Three edge-tilings. On each edge each median is affine, so the three traces are explicit intervals depending piecewise-linearly on (l_i, h_i) through $a_i = |I_i \cap [0, \frac{1}{2}]|$ and $b_i = |I_i \cap [\frac{1}{2}, 1]|$. Classifying each I_i by crossing type (contained in $[0, \frac{1}{2}]$, crossing $\frac{1}{2}$, or contained in $[\frac{1}{2}, 1]$) and each edge by the left-to-right order of its traces, the abutting (tiling) conditions become a finite family of linear systems in (l_i, h_i) .

Lemma 5.2. *The edge-tiling conditions have exactly three solutions with $0 \leq l_i < h_i \leq 1$: $[\frac{1}{3}, \frac{2}{3}]^3$, $[0, \frac{1}{3}]^3$, and $[\frac{2}{3}, 1]^3$.*

This is the case $\tau_i = \frac{1}{2}$ of Proposition A.1, proved in full in Appendix A for arbitrary cyclic triples: the symmetries of the configuration reduce the a priori 27 crossing patterns to seven classes, five of which are impossible and two of which lead to cyclic linear systems with unique solutions. The lemma has also been verified by three independent exact computer enumerations, archived as ancillary files.

Interior witness. The centroid G has $m(G) = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Since $\frac{1}{2} \notin [0, \frac{1}{3}] \cup [\frac{2}{3}, 1]$, the tilings $[0, \frac{1}{3}]^3$ and $[\frac{2}{3}, 1]^3$ leave G uncovered; they tile the boundary but are not coverings of Δ . The tiling $[\frac{1}{3}, \frac{2}{3}]^3$ covers Δ and has $\sum r_i = 1$. Hence the unique covering with $\sum r_i = 1$ is $[\frac{1}{3}, \frac{2}{3}]^3$. $\square$

6. A DIMENSION-AGNOSTIC NECESSARY CONDITION FOR TRANSPORT

The proofs of Theorems 3.1 and 5.1 share one device: a probability measure whose pushforward under each direction is uniform. Which direction-tuples admit one?

Proposition 6.1. *Let $u_1, \dots, u_{d+1}$ be normalized affine forms on Δ^d with vertex-value matrix V invertible. If a probability measure μ on Δ^d has $u_{i\#}\mu = \text{Leb}[0, 1]$ for all i , then its barycenter $p \in \Delta^d$ satisfies $Vp = \frac{1}{2}\mathbf{1}$; equivalently the mid-level hyperplanes $\{u_i = \frac{1}{2}\}$ concur at p , which requires $\mathbf{1}^\top V^{-1}\mathbf{1} = 2$ and $V^{-1}\mathbf{1} \geq 0$.*

Proof. $\frac{1}{2} = \mathbb{E}_\mu[u_i] = \sum_j V_{ij} \mathbb{E}_\mu[x_j] = (Vp)_i$, and $p \in \Delta^d$ gives $\mathbf{1}^\top p = 1$, $p \geq 0$; substitute $p = \frac{1}{2}V^{-1}\mathbf{1}$. $\square$

The three medians satisfy this with p the centroid; the three facets do not ($\mathbf{1}^\top V^{-1}\mathbf{1} = 3 \neq 2$), recovering Gardner's obstruction. The concurrence family is two-dimensional (parametrized by p): a normalized direction has vertex values $\{0, \tau, 1\}$, and $Vp = \frac{1}{2}\mathbf{1}$ ties each τ_i to p . The medians are the case $p = \text{centroid}$, forcing $\tau_i = \frac{1}{2}$. Among these, the explicit perimeter measure singles out the medians:

Proposition 6.2. *For a direction u with vertex values $\{0, \tau, 1\}$, the perimeter measure μ (uniform on each edge, mass $\frac{1}{3}$) satisfies $u_{\#}\mu = \text{Leb}[0, 1]$ if and only if $\tau = \frac{1}{2}$.*

Proof. On the three edges u is affine between the pairs $\{0, \tau\}, \{\tau, 1\}, \{0, 1\}$, so $u_{\#}\mu$ has density $\frac{1}{3\tau} + \frac{1}{3}$ on $[0, \tau]$ and $\frac{1}{3(1-\tau)} + \frac{1}{3}$ on $[\tau, 1]$. Both equal 1 iff $\tau = \frac{1}{2}$. $\square$

Uniform edge-weights thus single out the medians. Allowing *unequal* edge masses, however, the construction covers a full two-parameter family:

Theorem 6.3 (weighted perimeter). *Let $p = (p_A, p_B, p_C)$ be interior with $\max_j p_j < \frac{1}{2}$, and let $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ (vertex values at A, B, C) be the cyclic directions concurrent at p , i.e. $\tau_1 = \frac{1/2-p_C}{p_A}$, $\tau_2 = \frac{1/2-p_B}{p_C}$, $\tau_3 = \frac{1/2-p_A}{p_B}$. Let μ_p be uniform on each edge with masses $w_{AB} = 1 - 2p_C$, $w_{BC} = 1 - 2p_A$, $w_{CA} = 1 - 2p_B$ (summing to 1). Then $u_{i\#}\mu_p = \text{Leb}[0, 1]$ for $i = 1, 2, 3$; consequently every covering of Δ by planks in these three directions satisfies $\sum \text{rw} \geq 1$.*

Proof. Each u_i is affine of full range $[0, 1]$ on the edge joining its 0- and 1-vertices (u_1 on BC , u_2 on AB , u_3 on CA) and affine of range $[0, \tau_i]$, resp. $[\tau_i, 1]$, on the other two. The pushforward of a uniform edge mass w_e through an affine map of range length L is uniform with density w_e/L . Hence $u_{1\#}\mu_p$ has density $w_{BC} + w_{AB}/\tau_1$ on $[0, \tau_1]$ and $w_{BC} + w_{CA}/(1 - \tau_1)$ on $[\tau_1, 1]$, and similarly for u_2, u_3 : six equations. Using $p_A + p_B + p_C = 1$, each composite term collapses by the identity $(1 - 2x)p/(\frac{1}{2} - x) = 2p$: e.g. $w_{AB}/\tau_1 = (1 - 2p_C)p_A/(\frac{1}{2} - p_C) = 2p_A$, so $w_{BC} + w_{AB}/\tau_1 = (1 - 2p_A) + 2p_A = 1$; likewise $1 - \tau_1 = \frac{1/2-p_B}{p_A}$ gives $w_{CA}/(1 - \tau_1) = 2p_A$. All six densities equal 1, so each marginal is $\text{Leb}[0, 1]$, and the transport argument of Theorem 5.1 applies verbatim. Finally $\tau_i \in (0, 1)$ for all i iff $\max_j p_j < \frac{1}{2}$ iff all three masses are positive, so every valid cyclic triple admits μ_p . $\square$

Taking p the centroid recovers Theorem 5.1's bound ($\tau_i = \frac{1}{2}$, equal masses). For p off the centroid the directions are generic non-median triples; e.g. $p = (\frac{9}{20}, \frac{3}{10}, \frac{1}{4})$ gives $\tau = (\frac{5}{9}, \frac{4}{5}, \frac{1}{6})$ with masses $(\frac{1}{2}, \frac{1}{10}, \frac{2}{5})$. Theorem 6.3 thus proves the bound of Conjecture 1.1 for a two-parameter family of generic direction triples on the triangle, answering the finite-direction existence question of [5] affirmatively for the cyclic concurrent family.

Combining necessity (Proposition 6.1) with the explicit construction (Theorem 6.3) yields the first *characterization* in this setting:

Corollary 6.4. *Call a direction triple cyclic if its vertex values are $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ with $\tau_i \in (0, 1)$. A cyclic triple admits a uniform-marginal witness measure if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ concur. In that case the concurrence point p automatically lies in the open medial triangle ($p_j > 0$ and $p_j < \frac{1}{2}$ for all j), and the weighted-perimeter measure μ_p of Theorem 6.3 is a witness.*

Proof. The vertex-value matrix has $\det V = -(1 + \tau_1\tau_2\tau_3) \neq 0$, so Proposition 6.1 applies and gives necessity. Conversely, if the lines concur at p (a point of the affine plane of Δ , so $\sum_j p_j = 1$), then p is the unique solution of $Vp = \frac{1}{2}\mathbf{1}$, which in closed form reads $2(1 + \tau_1\tau_2\tau_3)p_A = 1 - \tau_3(1 - \tau_2)$ and cyclically; each right-hand side is positive for $\tau_i \in (0, 1)$, so $p_j > 0$, and the identities $\frac{1}{2} - p_C = \tau_1 p_A$, $\frac{1}{2} - p_A = \tau_3 p_B$, $\frac{1}{2} - p_B = \tau_2 p_C$ give $p_j < \frac{1}{2}$. Theorem 6.3 then supplies μ_p . $\square$

Remark 6.5. The *anti-cyclic* pattern (all 0- and 1-roles exchanged) reduces to the cyclic one by a reflection of Δ (a transposition of two vertices), which maps the medial triangle to itself; Theorem 6.3, Corollary 6.4 and Theorem 6.6 below hold for it verbatim. All remaining role assignments are classified by Theorem 6.8 below.

The bound of Theorem 6.3 is in fact *attained* at every member of the family, and the tight covering is unique — a two-parameter family of sharp configurations extending the median rigidity of §5.

Theorem 6.6 (tightness and rigidity of the cyclic family). *Let p and u_1, u_2, u_3 be as in Theorem 6.3. Write $\alpha = 1 - 2p_A$, $\beta = 1 - 2p_B$, $\gamma = 1 - 2p_C$ (the edge masses of μ_p ; $\alpha, \beta, \gamma > 0$, $\alpha + \beta + \gamma = 1$) and $S = \alpha\beta + \beta\gamma + \gamma\alpha$. Then the planks $P_i = \{u_i \in I_i\}$ with*

$$I_1 = \left[\frac{\alpha\gamma}{S}, 1 - \frac{\alpha\beta}{S} \right], \quad I_2 = \left[\frac{\beta\gamma}{S}, 1 - \frac{\alpha\gamma}{S} \right], \quad I_3 = \left[\frac{\alpha\beta}{S}, 1 - \frac{\beta\gamma}{S} \right]$$

cover Δ , with $\sum \text{rw} = \frac{\beta\gamma}{S} + \frac{\alpha\beta}{S} + \frac{\alpha\gamma}{S} = 1$. Moreover this is the unique covering of Δ with $\sum \text{rw} = 1$ by three planks of positive width, one in each of the directions u_1, u_2, u_3 . At p the centroid ($\alpha = \beta = \gamma = \frac{1}{3}$) it is $[\frac{1}{3}, \frac{2}{3}]^3$, recovering Theorem 5.1.

Proof. Write $l = (l_1, l_2, l_3) = (\alpha\gamma, \beta\gamma, \alpha\beta)/S$ and $h = (h_1, h_2, h_3) = \mathbf{1} - (\alpha\beta, \alpha\gamma, \beta\gamma)/S$, and note $\tau_1 = \frac{\gamma}{1-\alpha}$, $\tau_2 = \frac{\beta}{1-\gamma}$, $\tau_3 = \frac{\alpha}{1-\beta}$ from the formulas of Theorem 6.3. A direct substitution shows $(l, h) = (\lambda, \rho)$, the first solution of Proposition A.1; in particular $l_i < \tau_i < h_i$, and the traces of P_1, P_2, P_3 tile all three edges, so $\partial\Delta \subset \bigcup P_i$.

Step 1: an affine identity. Let $a_1 = \alpha(1 - \alpha)$, $a_2 = \gamma(1 - \gamma)$, $a_3 = \beta(1 - \beta)$ and $\Lambda = a_1u_1 + a_2u_2 + a_3u_3$. At the vertices: $\Lambda(A) = a_1\tau_1 + a_3 = \alpha\gamma + \beta(\alpha + \gamma) = S$, $\Lambda(B) = a_2 + a_3\tau_3 = \gamma(\alpha + \beta) + \alpha\beta = S$, $\Lambda(C) = a_1 + a_2\tau_2 = \alpha(\beta + \gamma) + \beta\gamma = S$. An affine function agreeing at three affinely independent points is constant: $\Lambda \equiv S$ on the plane of Δ . Moreover $\sum_i a_i = 1 - (\alpha^2 + \beta^2 + \gamma^2) = 2S$, and expanding (using $1 - \alpha = \beta + \gamma$ etc.)

$$\sum_i a_i l_i = \frac{\alpha^2\gamma(\beta+\gamma) + \beta\gamma^2(\alpha+\beta) + \alpha\beta^2(\gamma+\alpha)}{S} = \frac{(\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2) + \alpha\beta\gamma}{S} = \frac{S^2 - \alpha\beta\gamma}{S},$$

since $S^2 = \sum \alpha^2\beta^2 + 2\alpha\beta\gamma$; the same expansion gives $\sum_i a_i(1 - h_i) = \frac{S^2 - \alpha\beta\gamma}{S}$, whence $\sum_i a_i h_i = 2S - \frac{S^2 - \alpha\beta\gamma}{S} = S + \frac{\alpha\beta\gamma}{S}$. Thus

$$\sum_i a_i l_i = S - \frac{\alpha\beta\gamma}{S} < S < S + \frac{\alpha\beta\gamma}{S} = \sum_i a_i h_i. \quad (1)$$

Step 2: the planks cover Δ . Suppose not; the uncovered set U is relatively open (planks are closed) and misses $\partial\Delta$, and it decomposes as $U = \bigcup_\sigma P_\sigma$ over the eight sign patterns $\sigma \in \{\text{lo}, \text{hi}\}^3$, where $P_\sigma = H_\sigma \cap \text{int } \Delta$ and H_σ is the open convex set $\{u_i < l_i \text{ } (\sigma_i = \text{lo}), u_i > h_i \text{ } (\sigma_i = \text{hi})\}$; write c_i for the active bound (l_i or h_i). We show each P_σ is empty.

First, $\overline{P_\sigma} \cap \partial\Delta$ is finite: such a point x satisfies the closed pattern constraints, and is covered by some P_j (the boundary is tiled), forcing $u_j(x) = c_j$; each u_j is nonconstant on each edge (its vertex values there are distinct), so x lies in a set of at most nine points. Next, $H_\sigma \subseteq \overline{\Delta}$ whenever $P_\sigma \neq \emptyset$: otherwise pick $y \in H_\sigma \setminus \overline{\Delta}$ and a disk $B \subset P_\sigma$; for each $z \in B$ the segment $[z, y] \subset H_\sigma$ crosses $\partial\Delta$ at a point of $\overline{P_\sigma} \cap \partial\Delta$, and segments through the fixed y meeting a common crossing point have collinear z , so a disk of z 's produces infinitely many crossing points — contradicting finiteness. In particular H_σ is bounded.

No two of the lines $\ell_i = \{u_i = c_i\}$ are parallel: $u_i \parallel u_j$ would force $\tau_1(1 - \tau_2) = 1$, $\tau_2(1 - \tau_3) = 1$ or $\tau_3(1 - \tau_1) = 1$, impossible for $\tau \in (0, 1)^3$. If the three lines are concurrent, H_σ is an intersection of three open half-planes whose boundaries pass through one point, hence empty or an unbounded open cone; boundedness gives $H_\sigma = \emptyset$. Otherwise H_σ , if nonempty, is the open triangle with vertices $v_{ij} = \ell_i \cap \ell_j \in \overline{H_\sigma} \subseteq \overline{\Delta}$, and each v_{ij} satisfies the closed constraint in the third index k . Since $\Lambda \equiv S$, $u_k(v_{ij}) = (S - a_i c_i - a_j c_j)/a_k$, so that constraint reads $S \leq \sum_m a_m c_m$ if $\sigma_k = \text{lo}$, and $S \geq \sum_m a_m c_m$ if $\sigma_k = \text{hi}$. For a *mixed* pattern both directions occur, forcing $S = \sum_m a_m c_m$; but then $u_k(v_{ij}) = c_k$, i.e. the three lines concur — excluded. For $\sigma = (\text{lo}, \text{lo}, \text{lo})$ we get $S \leq \sum a_i l_i$, and for $\sigma = (\text{hi}, \text{hi}, \text{hi})$, $S \geq \sum a_i h_i$ — both contradict (1). Hence $U = \emptyset$: the planks cover, and $\sum \text{rw} = 1$.

Step 3: uniqueness. Let (I'_i) be any covering with all $r'_i = |I'_i| > 0$ and $\sum r'_i = 1$. Since $u_{i\#}\mu_p = \text{Leb}[0, 1]$ (Theorem 6.3), $\mu_p(P'_i) = r'_i$, and $1 = \mu_p(\Delta) \leq \sum \mu_p(P'_i) = 1$ forces the planks to partition μ_p a.e.; as μ_p has positive uniform density on each edge, the traces tile all three

edges. By Proposition A.1, (I'_i) is $([\lambda_i, \rho_i]) = (I_i)$, $([0, \lambda_i])$ or $([\rho_i, 1])$. The last two do not cover Δ : set $\delta = \frac{\alpha\beta\gamma}{2S^2}$ and let x^* be the unique point of the plane with $u_i(x^*) = \lambda_i + \delta$ for all i (consistent with $\Lambda \equiv S$ by (1) and $\sum a_i = 2S$; unique since V is invertible). Its barycentric coordinates are

$$x^* = \frac{1}{2S^2} \left(\alpha\beta(1-\alpha)(1+\alpha-\beta), \beta\gamma(1-\beta)(\alpha+2\beta), \alpha\gamma(\alpha+\beta)(2-2\alpha-\beta) \right),$$

all positive, so $x^* \in \text{int } \Delta$ and $u_i(x^*) > \lambda_i$ for every i : x^* is uncovered by $([0, \lambda_i])$. Symmetrically the point y^* with $u_i(y^*) = \rho_i - \delta$, with coordinates $\frac{1}{2S^2} (\alpha\gamma(1-\alpha)(2\alpha+\beta), \alpha\beta(1-\beta)(1+\beta-\alpha), \beta\gamma(\alpha+\beta)(2-\alpha-2\beta))$, is uncovered by $([\rho_i, 1])$. Hence $(I'_i) = (I_i)$. $\square$

Remark 6.7. At the centroid, $\delta = \frac{1}{6}$ and both witnesses degenerate to the point with $u_i \equiv \frac{1}{2}$ — the centroid itself, recovering the interior witness used in the proof of Theorem 5.1. The identities of Step 1 and the positivity of the witness coordinates, as well as exact coverage checks at several family points, are verified symbolically in the archived ancillary script `cyclic_family_tight_rigidity.py`.

Finally, the restriction to cyclic role assignments can be removed altogether. The plank problem is invariant under the *flip* of a normalized direction, $u \mapsto 1-u$, $I \mapsto 1-I$ (the planks, their widths, and uniformity of marginals are unchanged, and each mid-level line is fixed). Call the edge joining the 0- and 1-vertices of a direction its *full edge*. Flips reduce every role assignment to one of three shapes, and the existence question is settled in each:

Theorem 6.8 (three-direction characterization). *Let u_1, u_2, u_3 be normalized directions on Δ , pairwise non-parallel. A probability measure μ on Δ with $u_{i\#}\mu = \text{Leb}[0, 1]$ for $i = 1, 2, 3$ exists if and only if the three mid-level lines $\{u_i = \frac{1}{2}\}$ have a common point p . In that case $\sum \text{rw} \geq 1$ for every covering of Δ by planks parallel to these directions, and exactly one of the following holds:*

- (a) *the full edges are pairwise distinct: up to flips the triple is cyclic, p lies in the open medial triangle, and Theorems 6.3 and 6.6 apply verbatim (explicit witness μ_p ; unique tight covering);*
- (b) *all three directions have the same full edge e : then p is the midpoint of e , the uniform measure on e is a witness, and the bound follows from covering e alone (the trivial one-dimensional case recorded in Theorem 2.1).*

Proof. Necessity. If μ exists, its barycenter $p \in \Delta$ satisfies $u_i(p) = \mathbb{E}_\mu[u_i] = \frac{1}{2}$ for every i (no invertibility of V is needed), so the mid-lines concur at p .

Classification. (1) If the three full edges are distinct, flipping each direction so that its $(0, 1)$ -pair is oriented cyclically makes the triple cyclic; by Corollary 6.4, concurrence then places p in the open medial triangle and yields μ_p , and Theorem 6.6 the unique tight covering. (2) Suppose exactly two directions, say u_i and u_j , share their full edge e , and let x_v be the barycentric coordinate of the vertex v opposite e . After a flip, u_i and u_j agree on e , so $u_i - u_j = (\tau_i - \tau_j)x_v$ with $\tau_i \neq \tau_j$ (equality would make them parallel). Hence $\{u_i = \frac{1}{2}\} \cap \{u_j = \frac{1}{2}\} \subset \{x_v = 0\}$, the line through e , on which both mid-lines pass only through the midpoint of e ; but the third direction is not full on e , its values at the endpoints of e being $\{0, \tau_k\}$ or $\{\tau_k, 1\}$, so its value at the midpoint is $\frac{\tau_k}{2}$ or $\frac{1+\tau_k}{2}$, never $\frac{1}{2}$. Thus the mid-lines of such a triple *never* concur; consistently, no witness measure exists ($\mathbb{E}_\mu[u_i] = \mathbb{E}_\mu[u_j] = \frac{1}{2}$ would force $\mathbb{E}_\mu[x_v] = 0$, hence $\text{supp } \mu \subset e$, on which u_k is not onto $[0, 1]$). This shape contributes to neither side of the equivalence. (3) If all three directions share the full edge e , then after flips each restricts to e as the identity parameter, so all mid-lines pass through the midpoint of e (concurrence always holds), and the uniform probability measure on e has all three pushforwards equal to $\text{Leb}[0, 1]$. For the

bound, a covering of Δ covers e , on which the i -th plank traces an interval of length r_i ; hence $\sum r_i \geq 1$. $\square$

Remark 6.9. Theorem 6.8 answers the existence question of [5] completely for three directions on the triangle. In particular the open case of Conjecture 1.1 on the triangle is reduced to *non-concurrent* triples, for which no single witness measure can exist.

The concurrence condition is dimension-agnostic: the cyclic medians of Δ^d ($m_i(e_i) = \frac{1}{2}$, $m_i(e_{i+1}) = 0$, $m_i(e_{i-1}) = 1$, else $\frac{1}{2}$) have row-sums $\frac{d+1}{2}$, hence concur at the centroid in every dimension. Whether a witness measure then exists for $d \geq 3$ is open (the perimeter measure is special to $d = 2$); this is a higher-dimensional instance of the finite-direction question left open by Gardner [5].

7. A NORMALIZER OBSTRUCTION

For a direction u let $\ell_K(u)$ be the length of the longest chord of K parallel to u ; always $\ell_K(u) \leq w_K(u)$. Bakaev–Yehudayoff’s chord bound $\sum_i w(P_i)/\ell_K(u_i) \geq 1$ [4] follows from Bang’s sign argument via the chord lemma. Consider interpolating the normalizer, $N_c = (1 - c)\ell + c w$, $c \in [0, 1]$: $N_c = \ell$ at $c = 0$ (chord bound) and $N_c = w$ at $c = 1$ (Conjecture 1.1).

Proposition 7.1. *The chord-lemma step is sharp: for u with $\|u\| = a/2$, the body $(K - u) \cap (K + u)$ contains a homothet of K of ratio $(\ell - a)/\ell$ [6, Lem. 2.3], and this ratio is attained (by a segment). Consequently the iterated chord-lemma placement of a Bang-system witness $x_0 + \sum_j \pm u_j$ proves $\sum_i w(P_i)/N(u_i) \geq 1$ for $N = \ell$ and for no direction-normalizer $N > \ell$; in particular it does not establish $\sum_i w(P_i)/N_c(u_i) \geq 1$ for any $c > 0$.*

Proof. The placement argument runs as follows: iterating the chord lemma over the directions u_j , the common intersection $\bigcap_j ((K - u_j) \cap (K + u_j))$ (in the appropriate nested form) is nonempty as soon as $\sum_j w(P_j)/\ell_j < 1$, which places the witness and forces $\sum w(P_j)/\ell_j \geq 1$ for coverings. The movement budget available in direction u is therefore governed by the homothety ratio $(\ell - a)/\ell$. This ratio is tight: for a segment $K = [y, z]$ of length ℓ parallel to u , $(K - u) \cap (K + u)$ is a segment of length exactly $\ell - a$. Hence the total budget in direction u is exactly the longest chord $\ell(u)$ and cannot be stretched to any $N(u) > \ell(u)$; since $N_c(u) = (1 - c)\ell(u) + c w_K(u) > \ell(u)$ whenever $c > 0$ and $w_K(u) > \ell(u)$, the claim follows. $\square$

Remark 7.2. Proposition 7.1 does *not* show $\sum w(P_i)/N_c(u_i) \geq 1$ is false for $c > 0$; it shows the sign-argument route cannot prove it. Beating $2/(1 + \sqrt{d})$ by this normalizer therefore requires a witness that is not a Bang-system point, i.e. one exploiting the coupling of the covering. This is consistent with the sharpness of [4] for the cube.

8. OPEN PROBLEMS

Problem 8.1. The three-plank case for three genuinely tilted directions on the triangle (equivalently Conjecture 1.1 for the triangle). By Theorem 6.8 the remaining case is that of *non-concurrent* triples, where no single witness measure exists.

Problem 8.2. Theorem 6.8 settles the existence of uniform-marginal witness measures for three directions on the triangle. For $d \geq 3$ the concurrence condition of Proposition 6.1 remains necessary; is it sufficient for $d + 1$ directions on Δ^d ? The first test case is the cyclic medians of Δ^d , which concur at the centroid in every dimension (§6).

Problem 8.3. Whether $\sum w(P_i)/N_c(u_i) \geq 1$ holds for some $c > 0$ (a coupling certificate), which would improve the general bound past $2/(1 + \sqrt{d})$.

APPENDIX A. EDGE TILINGS OF CYCLIC TRIPLES

Throughout, $u_1 = (\tau_1, 0, 1)$, $u_2 = (0, 1, \tau_2)$, $u_3 = (1, \tau_3, 0)$ is a cyclic triple with arbitrary $\tau_1, \tau_2, \tau_3 \in (0, 1)$ (no concurrence is assumed), and $I_i = [l_i, h_i] \subset [0, 1]$, $l_i < h_i$, are three intervals. Say (I_1, I_2, I_3) satisfies the *edge-tiling conditions* if on each edge of Δ the traces $T_i = \{x \in \text{edge} : u_i(x) \in I_i\}$ cover the edge and have pairwise disjoint interiors. Indices are mod 3.

Proposition A.1. *For every $\tau \in (0, 1)^3$ the edge-tiling conditions have exactly three solutions: with*

$$\lambda_i = \frac{\tau_i(1 - \tau_{i+1}(1 - \tau_{i+2}))}{1 + \tau_1\tau_2\tau_3}, \quad \rho_i = \frac{1 - \tau_{i+2}(1 - \tau_i)}{1 + (1 - \tau_1)(1 - \tau_2)(1 - \tau_3)},$$

they are $([\lambda_i, \rho_i])_i$, $([0, \lambda_i])_i$, and $([\rho_i, 1])_i$. One has $0 < \lambda_i < \tau_i < \rho_i < 1$ for all i and all τ . For $\tau_i \equiv \frac{1}{2}$ (the medians) the solutions are $[\frac{1}{3}, \frac{2}{3}]^3$, $[0, \frac{1}{3}]^3$, $[\frac{2}{3}, 1]^3$.

Setup. Parametrize the edges by $t \in [0, 1]$ in the orientations $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow A$. From the vertex values, each edge sees one direction with values $[0, \tau_c]$ (the *low* direction c , $u_c(t) = \tau_c(1 - t)$), one with values $[0, 1]$ (the *full* direction f , $u_f(t) = t$), and one with values $[\tau_g, 1]$ (the *high* direction g , $u_g(t) = 1 - (1 - \tau_g)t$), with roles

$$(c, f, g) = (1, 2, 3) \text{ on } AB, \quad (3, 1, 2) \text{ on } BC, \quad (2, 3, 1) \text{ on } CA.$$

Inverting, the traces on an edge with roles (c, f, g) are

$$\begin{aligned} T_c &= \left[\max\left(0, 1 - \frac{h_c}{\tau_c}\right), 1 - \frac{l_c}{\tau_c} \right] \quad (\text{nondegenerate iff } l_c < \tau_c), & T_f &= [l_f, h_f], \\ T_g &= \left[\frac{1 - h_g}{1 - \tau_g}, \min\left(1, \frac{1 - l_g}{1 - \tau_g}\right) \right] \quad (\text{nondegenerate iff } h_g > \tau_g). \end{aligned} \tag{2}$$

Classify each I_i by its position relative to τ_i : type L if $h_i \leq \tau_i$, type R if $l_i \geq \tau_i$, type M if $l_i < \tau_i < h_i$; every I_i is of exactly one type. Two elementary observations, used throughout:

(O1) If I_c is of type M then $T_c = [0, 1 - l_c/\tau_c]$ (it contains the left endpoint); if I_c is of type R then T_c is degenerate (a point or empty). Dually, if I_g is of type M then $T_g = [(1 - h_g)/(1 - \tau_g), 1]$; if I_g is of type L then T_g is degenerate.

(O2) The tiling condition is insensitive to degenerate traces: a degenerate trace covers at most one point, and since the nondegenerate traces are finitely many closed sets whose union contains $[0, 1]$ minus finitely many points, that union is all of $[0, 1]$. So on each edge the *nondegenerate* traces tile $[0, 1]$; in particular, finitely many closed intervals of positive length with disjoint interiors covering $[0, 1]$ abut consecutively from 0 to 1.

Symmetries. The rotation $A \rightarrow B \rightarrow C \rightarrow A$ maps the cyclic triple with parameters (τ_1, τ_2, τ_3) to the cyclic triple with parameters (τ_2, τ_3, τ_1) and acts on solutions by $(I_1, I_2, I_3) \mapsto (I_2, I_3, I_1)$, shifting type vectors cyclically. The reflection swapping B and C , composed with the flip $s \mapsto 1 - s$ of each normalized direction (which leaves every plank unchanged), maps the cyclic triple (τ_1, τ_2, τ_3) to the cyclic triple $(1 - \tau_1, 1 - \tau_3, 1 - \tau_2)$ and acts by $(I_1, I_2, I_3) \mapsto (1 - I_1, 1 - I_3, 1 - I_2)$ (where $1 - [l, h] = [1 - h, 1 - l]$), exchanging types L $\leftrightarrow$ R. Since we prove Proposition A.1 for *all* τ at once, these symmetries permute the class of problems, and the 27 type vectors $(X_1, X_2, X_3) \in \{L, M, R\}^3$ reduce to seven orbit representatives:

$$\text{MMM}; \quad \text{LLL} (\sim \text{RRR}); \quad \text{MML}; \quad \text{LLM}; \quad \text{LLR}; \quad \text{LMR}; \quad \text{LRM}$$

(orbit sizes 1, 2, 6, 6, 6, 3, 3). It suffices to analyze one representative per orbit.

Case MMM. By (O1), on every edge $T_c = [0, 1 - l_c/\tau_c]$ and $T_g = [(1 - h_g)/(1 - \tau_g), 1]$, both nondegenerate, and T_f has positive length. Disjointness of interiors forces T_f inside the gap between them, and covering forces the abutments

$$l_f = 1 - \frac{l_c}{\tau_c}, \quad h_f = \frac{1 - h_g}{1 - \tau_g}.$$

Reading this on the three edges gives two decoupled cyclic systems,

$$l_2 = 1 - \frac{l_1}{\tau_1}, \quad l_1 = 1 - \frac{l_3}{\tau_3}, \quad l_3 = 1 - \frac{l_2}{\tau_2}; \quad h_1 = \frac{1 - h_2}{1 - \tau_2}, \quad h_2 = \frac{1 - h_3}{1 - \tau_3}, \quad h_3 = \frac{1 - h_1}{1 - \tau_1}.$$

Each system composes three decreasing affine maps, so the composite self-map (of l_1 , resp. h_1) is affine with negative slope $(-1/(\tau_1\tau_2\tau_3))$, resp. $-1/\prod(1 - \tau_i)$, hence has a unique fixed point: the solutions are $l_i = \lambda_i$, $h_i = \rho_i$ (direct substitution verifies the formulas; e.g. $1 - \lambda_1/\tau_1 = (\tau_1\tau_2\tau_3 + \tau_2(1 - \tau_3))/(1 + \tau_1\tau_2\tau_3) = \lambda_2$). The chain $0 < \lambda_i < \tau_i < \rho_i < 1$ reduces to $\tau_{i+1}(\tau_{i+2} - 1) < \tau_1\tau_2\tau_3$ and $\tau_i(1 - \tau_{i+1}) < 1$, both automatic; so the solution is of type MMM and, by construction, tiles each edge as $[0, l_f] \cup [l_f, h_f] \cup [h_f, 1]$.

Case LLL. All T_g are degenerate, so each edge is tiled by $T_c = [1 - h_c/\tau_c, 1 - l_c/\tau_c]$ (the full image, as $I_c \subset [0, \tau_c]$) and $T_f = I_f$ with $h_f \leq \tau_f < 1$. The tile containing 1 must be T_c , forcing $l_c = 0$. If $h_c = \tau_c$ then $T_c = [0, 1]$ and T_f has empty interior, impossible; so T_f contains 0 ($l_f = 0$) and abuts T_c : $h_f = 1 - h_c/\tau_c$. On the three edges: $l_i = 0$ for all i , and the h_i satisfy the *same* cyclic system as the l_i in case MMM, so $h_i = \lambda_i$, giving $([0, \lambda_i])_i$ (type L since $\lambda_i < \tau_i$). Applying the reflection symmetry, the orbit representative RRR yields the unique solution $h_i = 1$, $l_i = \rho_i$, i.e. $([\rho_i, 1])_i$; equivalently, the l_i satisfy the same cyclic system as the h_i in case MMM.

Case MML (impossible). On AB : T_3 is degenerate (I_3 of type L), so $[0, 1]$ is tiled by $T_1 = [0, 1 - l_1/\tau_1]$ (O1) and $T_2 = I_2$. If $1 \in T_1$ then $T_1 = [0, 1]$ and T_2 has empty interior, impossible; so $h_2 = 1$. Then on BC the high trace (2) is $T_2 = [\frac{1-h_2}{1-\tau_2}, \min(1, \frac{1-l_2}{1-\tau_2})] = [0, 1]$ (as $h_2 = 1$ and $l_2 < \tau_2$), leaving the full trace $T_1 = I_1$ (positive length) with empty interior — contradiction.

Case LLM (impossible). On BC $((c, f, g) = (3, 1, 2))$: the high trace is degenerate (I_2 of type L), so $[0, 1]$ is tiled by $T_3 = [0, 1 - l_3/\tau_3]$ (O1, I_3 of type M) and $T_1 = I_1$ with $h_1 \leq \tau_1 < 1$. The tile containing 1 must be T_3 , so $l_3 = 0$ and $T_3 = [0, 1]$, leaving T_1 with empty interior — contradiction.

Case LLR (impossible). On BC both compressed traces are degenerate (I_3 of type R gives a degenerate low trace, I_2 of type L a degenerate high trace), so $T_1 = I_1$ alone tiles $[0, 1]$, forcing $h_1 = 1 > \tau_1$ — contradicting type L.

Case LMR (impossible). On BC : low trace degenerate (I_3 type R), so $T_1 = I_1$ and $T_2 = [\frac{1-h_2}{1-\tau_2}, 1]$ (O1) tile $[0, 1]$; as above this forces $l_1 = 0$ and $h_1 = \frac{1-h_2}{1-\tau_2}$. On CA : high trace degenerate (I_1 type L), so $T_2 = [0, 1 - l_2/\tau_2]$ (O1) and $T_3 = I_3$ tile, forcing $h_3 = 1$ and $l_3 = 1 - l_2/\tau_2$. Now on AB the high trace is $T_3 = [0, \frac{1-l_3}{1-\tau_3}] = [0, \frac{l_2}{\tau_2(1-\tau_3)}]$ (truncated at 1 if larger, in which case T_2 already has empty interior, a contradiction), of positive length, so $l_2 > 0$; since $\tau_2(1 - \tau_3) < 1$, its right endpoint exceeds l_2 , and the interiors of T_3 and $T_2 = [l_2, h_2]$ overlap — contradiction.

Case LRM (impossible). On CA : both compressed traces are degenerate (I_2 of type R, I_1 of type L), so $T_3 = I_3$ alone tiles $[0, 1]$, i.e. $I_3 = [0, 1]$. Then on AB the high trace (2) is $T_3 = [\frac{1-h_3}{1-\tau_3}, \min(1, \frac{1-l_3}{1-\tau_3})] = [0, 1]$, leaving $T_2 = I_2$ (positive length) with empty interior — contradiction.

All orbits are exhausted, proving Proposition A.1. $\square$

The three solutions have also been verified by exact rational searches at $\tau = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$ and $\tau = (\frac{1}{2}, \frac{2}{3}, \frac{1}{3})$, and the median enumeration was verified by two further independent exact computations; all scripts are archived as ancillary files.

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